MORGAN C. FITZGERALD

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EDUCATION

Ph.D., NEUROSCIENCE; GPA 3.99 Anticipated advancement: 2025
UC, San Diego – San Diego, CA

B.S., BIOPSYCHOLOGY; GPA: 4.00 June 2020
UC, Santa Barbara – Santa Barbara, CA
- Senior Thesis: "Cerebellar network organization across the human menstrual cycle"

Vassar College – Poughkeepsie, NY August 2014—December 2016

RESEARCH EXPERIENCE

GRADUATE STUDENT RESEARCHER June 2023—Present
UC, San Diego, Department of Cognitive Science, Dr. Bradley Voytek

STAFF RESEARCH ASSOCIATE September 2020—August 2022
UC, San Diego Health, Department of Psychiatry, Dr. Matthew Panizzon
- Independent project examining sex-specific factors involved in migraine risk
- Assisted with writing and editing a NIH Research Project Grant (R01) centered on examining the effects of ovarian hormone suppression
- Operated MRI 3.0 T Siemens Prisma scanner
- Edited Institutional Review Board applications and wrote amendments for multiple studies
- Recruited subjects for NIH R01-funded studies
- Administered and scored neuropsychological batteries
- Utilized EPIC to conduct medical record chart reviews
- Constructed and managed databases for multiple studies

RESEARCH ASSISTANT March 2018—June 2020
UC, Santa Barbara, Department of Biopsychology, Dr. Emily Goard-Jacobs
- Investigated the effect of endogenous hormone fluctuations on the functional connectivity of the brain
- Processed fMRI data within MATLAB and RStudio
- Prepared blood samples to be sent for testing
- Assisted with MRI scans – loaded participants and ran task protocols
- Contributed to data collection, organization, and analysis across multiple studies

RESEARCH ASSISTANT September 2018—March 2019
UC, Santa Barbara, Department of Biopsychology, Dr. Scott Grafton
- Researched the efficacy of a novel micro-resonator in detecting impedance cardiography signal
- Monitored impedance cardiography and electrocardiogram waveforms during MRI scans
- Conducted ICG and ECG recordings on study participants
- Analyzed data using Moving Ensemble Analysis Pipeline (MEAP)

RESEARCH ASSISTANT August—December 2016
Vassar College, Department of Neuroscience, Dr. Bojana Zupan
- Researched the role of dopaminergic signaling within the amygdala in the acquisition of long-term depression and anxiety-like behaviors in rats
- Conducted rodent behavioral assays including open field, forced swim, and social defeat
- Completed bilateral stereotaxic injections into rat amygdala
- Performed transcranial perfusion of a rat and histological analysis on brain tissue

RESEARCH ASSISTANT August—December 2016

Vassar College, Department of Neuroscience, Dr. Kelli Duncan

- Investigated the neuroprotective response of steroidogenic enzymes in birds following traumatic brain injury
- Utilized quantitative polymerase chain reaction, RNA isolation and cDNA synthesis to quantify enzyme concentrations in brain tissue

PUBLICATIONS

- **Fitzgerald, M. C.,** Saelzler, U. G., & Panizzon, M. S. (2021). Sex Differences in Migraine: A Twin Study. *Frontiers in Pain Research*, *2*, 113.
    - **Media**:
        - Frontiers Science News (link), MD Edge (link), Physician's Weekly (link), Healthline (link)
- **Fitzgerald, M.,** Pritschet, L., Santander, T., Grafton, S. T., & Jacobs, E. G. (2020). Cerebellar network organization across the human menstrual cycle. *Scientific reports*, *10*(1), 20732.

POSTERS & PRESENTATIONS

- **Fitzgerald, M.**, (June 2022), '*Mechanisms of Sex Differences in Migraine: A Twin Study – The need to consider sex as a variable of interest'* [Presentation], American Headache Society, Denver, CO
- **Fitzgerald, M.**, Saelzler, U., & Panizzon, M., (May 2022), '*Hormonal Mechanisms of Sex Differences In Migraine*' [Poster], Organization for the Study of Sex Differences, Marina Del Rey, CA
- **Fitzgerald, M.**, (May 2020), '*Cerebellar network organization across the human menstrual cycle'* [Poster], Undergraduate Research and Creative Activities Conference, University of California, Santa Barbara

AWARDS & GRANTS

UC, San Diego

- Neuroscience Graduate Program NIH training grant (T32; 2023-2024)

UC, Santa Barbara

- **The Morgan Award for Research Promise** (2020)
    - Awarded to seniors who demonstrate promise in research, as selected by the faculty
- **University Award of Distinction** (2020)
    - Awarded to seniors who demonstrate in-depth involvement and significant achievement on campus
- **Exceptional Academic Performance** (2020)
    - Awarded to seniors with a 3.9 or higher GPA in their upper division major coursework
- **Distinction in the Major** (2020)
    - Awarded to seniors who successfully complete their senior thesis
- **Exceptional Student** (2019, 2020)
    - Awarded to students with a GPA of 3.75 or higher
- **Undergraduate Research and Creative Activities Grant** (2019)
    - Funding awarded to undergraduates to pursue an independent research project
- **Dean's Honors**, **College of Letters and Science** (2018, 2019, 2020)
    - Awarded to students with a GPA of 3.75 or higher

Vassar College

- **Dean's List** (2014, 2015, 2016)
    - Awarded to students with a GPA of 3.2 to 3.74
- **Liberty League All-Academic Team, Women's Tennis** (2016)
    - Awarded to students with a GPA above 3.2 and on a collegiate varsity sports team
- **Liberty League ITA Scholar-Athlete, Women's Tennis** (2016)
    - Awarded to students with a GPA above 3.2 and on a collegiate varsity sports team
- **All-Liberty League Honorable Mention, Women's Tennis** (2015-2016)
    - Awarded to the top all-conference athletes

WORK EXPERIENCE

RESEARCH ASSOCIATE September 2020— August 2022
UC, San Diego Health, Department of Psychiatry, Dr. Matthew Panizzon

- Described under 'Research Experience'

SCIENCE AND ENGINEERING LAB INTERN June–September 2015, 2016, 2017
Genentech – Oceanside, CA

- Planned and executed independent study for qualification of disposable spinners
- Maintained working cell lines for laboratory studies
- Monitored cell culture performance: gases, nutrients, metabolites, viable cell density, and pH
- Developed and introduced a stage-gate model for qualification of engineering and qualification runs
- Performed daily maintenance on Cedex BioHT and larger annual/biannual maintenance tasks
- Assisted with BioHT AOP development and documentation
- Prepared media per SOP for cell culture process
- Performed DO and pH calibrations for bioreactors
- Analyzed cell culture performance data using Delta-V and PI interface
- Developed a document database on TouchPoint
- Conducted root cause analysis of contamination

LEADERSHIP & ACTIVITIES

- **Student Representative/Policy Strategist**, Office of Student Affairs, UC San Diego May 2023 – Present
- **Lead Mentor/Advisory Committee,** Colors of the Brain, UC, San Diego September 2022 – Present
- **Co-Executive Director,** Miles for Migraine, San Diego December 2022 – Present
- **Patient Advocate,** Patients Rising, San Diego June 2021 – Present
- **Patient Advocate,** Global Healthy Living Foundation, San Diego, CA December 2020 – Present

- **Mentor**, Girls Inc. March 2019 – June 2021
- **Player**, Vassar College Women's Varsity Tennis Team, Poughkeepsie, NY August 2014 – December 2016
- **Member**, Women in STEM, Vassar College Science Club August 2014 – December 2016
- **Member**, Student Athletic Advisory Committee, Vassar College August 2015 – December 2016
- **Patient Companion**, Hospice of North Coast, San Diego, CA June 2016 – September 2016
- **Coach**, Special Olympics, San Diego, CA September 2011 – August 2014

LABORATORY SKILLS

- Administering neuropsychological batteries
- fMRI Scan Operator
- Transcardial perfusion
- Rodent/bird handling
- Cryostat
- Stereotaxic injection
- Free-floating immunohistochemistry
- Calcium imaging
- Real Time qPCR
- Good Manufacturing Practices (GMP)
- Antiseptic technique
- Cell culture maintenance
- cDNA synthesis

COMPUTER & LANGUAGE SKILLS

COMPUTER – Python**,** MATLAB, RStudio, Qualtrics, Microsoft Excel, PowerPoint, and Word

LANGUAGE – Spanish, basic written and verbal proficiency